

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Data Project Laboratory

Week 14 · Reproducibility, Multivariate Analysis and Statistical Communication

Tashkent International University · Faculty of Economics

2025/2026 academic
year



No new methods this week

What thirteen weeks have taught

Description · probability · distributions · sampling distribution · estimation · test · two samples · χ^2 · regression · inference · causality.

Now you have the tools. This week it's about turning them into one reliable report.

And one awkward question

If two analysts look for the same answer to the same question from the same data — will they arrive at the same answer?

Today we're going to measure it. The answer: most of the time, no one cheated.

Three parts of the week

1. Branching paths — one question, 48 well-founded analyses, and 23 of them are "significant."

2. Reproducibility — what breaks and how much: seedless interval, false positives, code that doesn't run.

3. Communication — what to write, what not to write, and in what order.

Common mistakes

Your project report is due this week. Each slide today corresponds to one item in the report. The last slide has a ten-point checklist — keep it with you as you write your report.

Today's Objectives

By the end of the session, you will be able to: 1. Create a set of all reasonable specifications for a single question and measure the sensitivity of the result to them (CLO8);

2. Write an estimate to the standard error of the mean (CLO5);

3. List the four mandatory parts of a repeatable design and implement them in your own design (CLO9);

4. State a single result with its conditions and constraints—without resorting to causal language (CLO7, CLO9);

5. Clearly document the use of SI and re-examine each of its numbers (CLO9).

This week's formulas and lists

F202 multivariate analysis F203
meaningful room rule F204
minimum repeatability F205
report structure (ten items) F206
SI statement (six steps) F207
sensitivity analysis

None of these are calculation formulas. All are decision rules. Therefore, they should be applied, not memorized.

Advance preparation

Gelman & Loken (2014) — 6 pages — should be read before the lecture. One of his phrases (“a garden of branching paths”) is the title of today's entire Section A.

Garden of Branching Paths

One question, five decisions, 48 honest answers

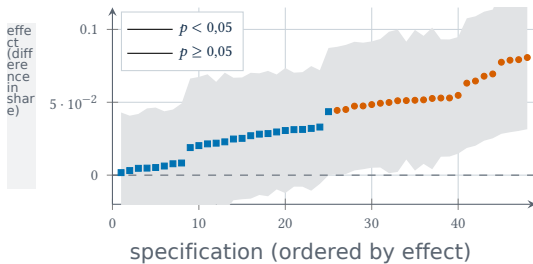
The question is “Do PG-13 movies make more money than R movies?”

The question seems straightforward. But it requires five decisions to answer, and each has a valid alternative.

Five decisions

1. What is “profitable”? Is the revenue more than 2.0 2.5 3.0 times the budget? 2. Do we exclude the 31 lines with zero revenue? (Week 12: they are a data error.) 3. Do we adjust for genre? (Week 13: genre is a confounding factor.) 4. Do we exclude the top 1 percent of budgets? 5. Do we add PG to PG-13?

$3 \times 2 \times 2 \times 2 \times 2 = 48$ specifications. None of them are wrong.

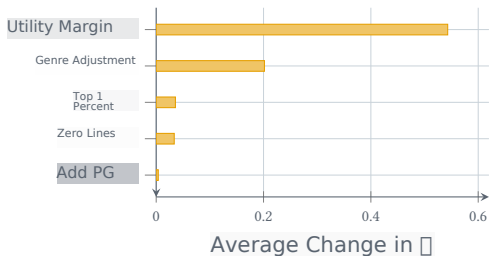


Result

$\square < 0.05$: 23 items · $\square \geq 0.05$: 25 items — almost a coin toss (0.479167).

Effect estimate: 0.001713 to 0.080708 — 47.12 times. $\square$: 0.001302 to 0.935178.

Which decision determines the outcome?



Decision $\Delta \square$

Utility Margin 0.543134
Genre Adjustment 0.201692
Remove Top 1 Percent 0.035834
Remove Zero Lines 0.033609
Add PG 0.003852

Largest Impact — Least Statistical Decision

What determines the outcome is the definition of the word “useful.” It’s not a statistical decision—it’s a substantive decision. There’s a 2.5x rule in cinema, but it’s not a law.

Therefore

In the report, the definition comes before the formula. If you do not write how you defined “useful film” and why you defined it that way, the reader will not be able to check your result.

And if you chose it after viewing the data — this is $\square$ -hacking from week 10, by another name.

Audience question

Question to the class. “Add PG” $\Delta \square = 0.003852$ — so this decision is insignificant? (In this data — yes; in other data it may be the most important. Sensitivity should be measured every time.)

Two honest solutions

Solution 1 — pre-recording (week 10)

Write down and record five decisions before you see the data.

Strength: Whatever the outcome, it is not your choice. Limitation: If you come up with a new, better idea after seeing the data — you label it as a search and bring it up separately.

Solution 2 — multivariate analysis (F202)

Calculate all the reasonable specifications and show the machine — like the graph above.

Strength: The reader can see for themselves how robust the result is. Limitation: You draw the line of “reasonable” anyway.

For your project — minimum requirement

A full multivariate analysis does not fit into a 2,500-word report. Therefore, the course requires a sensitivity analysis (F207):

Choose one of your main decisions, change it, rerun the analysis, and report the results.

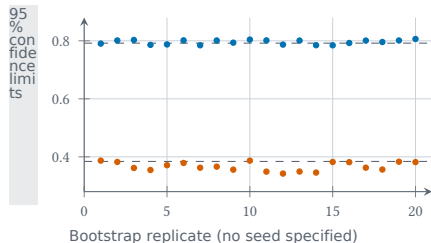
Two sentences are enough: “If we take the cutoff of 3.0 instead of 2.5, the difference drops from 0.0807 to 0.0017 and is no longer significant. So the result is sensitive to this decision.”

A common mistake

This does not make you look weak. On the contrary: a researcher who says that his result is sensitive to the decision is more credible. A researcher who hides this — when someone else finds out later — loses credibility.

Repeatability

Seed: single line of code, measured difference



Conditions Same data, same code, run 20 times; different seed each time. Solid lines — range [0.384127; 0.791940] (week 13).

Measured variance

Lower bound: 0.342324 to 0.386951 $\Rightarrow$ variance 0.044627
Upper bound: 0.784526 to 0.806331 $\Rightarrow$ variance 0.021805

Rounding to three places, the lower bound has 13 different values — meaning each run will produce a different report.

The seed does not lose randomness

It allows you to restore randomness. The result is still random — but it is the same random result.

`set.seed(20261401)` or the LCG seed — a single line. Without it, no one, not even you, can recreate your report.

How many rooms should be written? — the standard error says

Rule (F203)

The first meaningful room of the standard error is the last meaningful room of the estimate.

model $\pm$ 1 SE room

Kazakh. potatoes	0.588033	0.086232	3
Uzbek. wheat	0.134864	0.012480	3
Films budget	2.739510	0.056619	3

So not 0.588033, but 0.588 (or 0.59). The remaining rooms are noise.

A common mistake

But only IN CHOP. Full accuracy is maintained in the calculation. This is exactly the error that occurred in week 13: $44.52 - 30.35 = 14.17$, which is actually 14.1645 — rounded first and then subtracted.

Why this matters

Writing 0.588033 tells the reader, “I know this to six places.” This is a lie: the standard error is 0.086232, meaning nothing is known after the third place.

False precision is the most common and easiest reporting flaw to fix.

Independent verification is free and effective

Calculate the same result two ways.

With the open formula: $\pm$ 1 = 0.588033 ...
scipy.stats.linregress: $\pm$ 1 = 0.588033 ...
Difference: 0.00; also 0.00 in SE.

If it matches — the probability of a bug in the code is sharply reduced. If it doesn't match — you need to fix one, and you know it.

What is a recursive project (F204)



The toughest test — and it's free

Open a new, empty session. Run the code from start to finish once.

Is every number in the report re-generated?

Otherwise, your report is not a duplicate, and it's good that you found it, not someone else.

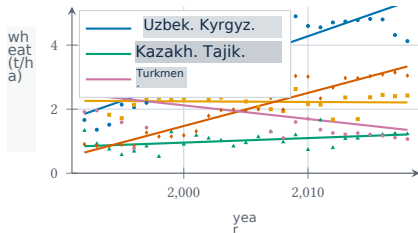
Why is this required in this course

Every number in the thirteen weeks of material this semester has been generated and checked by a script. Your project is required to do the same.

The reason is simple: a manually copied number will definitely have one mistake. Who finds it — you or the instructor — is the only difference.

Sample analysis

From question to conclusion: six steps



Step 1. Question

“Did wheat yield increase in Central Asia from 1992 to 2018, and is the increase uniform across countries?”
Variable, period, and population — all three are fixed.

Step 2. Data and its limitations

OWID/FAOSTAT · 135 rows · 5 countries × 27 years · 0 missing values. Limitation: this is aggregate data at the country level. Inference at the farm level is an ecological fallacy.

Stage 3-4. Description, model and CONDITIONS

country $\square$ 1 SE lag-1 Condition I

Uzbekistan	0.13486	0.01248	0.7897	violated
Kyrgyzstan	-0.00189	0.00698	0.1503	fulfilled
Kazakhstan	0.01411	0.00591	-0.2933	fulfilled
Tajikistan	0.10340	0.00584	0.1118	fulfilled
Turkmenistan	-0.04259	0.01849	0.6499	violated

The independence condition is violated in two of the five rows.

Steps 5-6: Conclusion considering the condition, and what was not proven

Step 5. Corrected conclusion (F199)

Country χ^2 eff ts. χ^2 decision

Uzbekistan 3.1726 0.138445 not rejected
Kyrgyzstan — 0.789240 not rejected Kazakhstan
— 0.024850 rejected Tajikistan — < 0.000001
rejected Turkmenistan 5.7301 0.352531 not
rejected

Before correction: Uzbekistan $\chi^2 = 6.556 \times 10^{-11}$, Turkmenistan $\chi^2 = 0.029882$.
Violation of the condition changed both conclusions — both can no longer be
rejected.

Step 6. What is NOT PROVEN — mandatory b

(a) No reason given. Seeds, irrigation, fertilizer, weather, politics — none of which are in the data. (b) Cross-country comparisons are full of confounding factors: climate, land, irrigation system not controlled for. (c) Ecological fallacy: no conclusion at the farmer level. (d) No prediction: extrapolation beyond 2018. (e) Adjusted χ^2 estimated, not precise.

Audience question

This section is the shortest and most valuable part of the report. It shows the reader that you can be trusted — because you know the limits of your work.

Communication

One result, four different statements

(1) and (2) are both bad

(1) "Productivity increased significantly ($p < 0.001$)."

No effect · no unit · no interval · no condition. The reader does not know how much it increased.

(2) " $\beta_1 = 0.103404$, $\beta = 1.152 \times 10^{-15}$, $\beta_2 = 0.926243$." Just numbers. The reader does not know the meaning and cannot find the unit.

(3) good, (4) best

(3) "In Tajikistan, wheat yield increased by an average of 0.103 t/ha per year from 1992 to 2018 (95% CI: 0.091–0.115; $p < 0.001$)."

(4) = (3) + "The lag-1 correlation of the residuals is 0.112, the threshold is 0.377 — the condition of independence is met, therefore, no correction is necessary. The reason is not given."

Audience question

The difference is not in length — (4) is only two sentences longer. The difference is in verifiability: the reader can recalculate (4) and know where the uncertainty lies.

Table or graph? And SI statement

Selection rule Graph: shape, trend, outlier, distribution, comparison. Table: when exact value is needed; when less than 3-4 rows; when the student needs to copy the number to his/her own account. Both: only if the result is central. Never: three-dimensional columns · truncated π -axis (week 1) · meaningless color · graph without legend.

SI statement (F206) — mandatory

“Used / how” for the six steps: research question · finding information · writing code · analyzing decisions · writing text · checking.

SI is not prohibited. Hiding is prohibited.

Three Hard Rules

1. Recalculate every number SI produces — SI's errors in weeks 10-13 were in the numbers. 2. Cite every source SI cites. Citing a nonexistent source is academic fraud. 3. Justify every analytical decision SI makes. “SI said so” is not a justification: today's Section A showed how a decision changes the outcome.

Conceptual test

Conceptual test — first individually, then in pairs

Question 1 The analyst calculated all 48 specifications and only reported the smallest β (0.001302). He did not lie about the others — he just wrote it. This is:

(A) honest, because no number is wrong (B) β -hacking (C) simple truncation

Answer: (B)

Each number is correct individually, but the selection process creates the result. If the student does not see 23 and 25, he cannot know the meaning of the β he sees. It is not the wrong number — it is the wrong picture. Concealment is also a form of cheating.

Question 2 $\beta_1 = 0.588033$, $SE(\beta_1) = 0.086232$. Which entry in the report is correct?

(A) 0.588033 (B) 0.59 (C) 0.6 (D) 0.5880

Answer: (B)

The first significant digit of SE is in hundredths (0.0862). Therefore, the estimate is also in hundredths (F203). (A) and (D) are falsely accurate; (C) loses the accuracy it provides.

Important: this is only a printing rule. Full accuracy is used when calculating the range.

Common error

Method (Mazur). 1 minute individual answer → find your neighbor who answered differently → 2 minutes convince him/her → vote again. If you have changed your mind, explain why.

**Summar
y**

Ten most common errors in reporting

1-5 1. Ineffective $\square$. There is " $\square < 0.001$ ", $\square$ 1 and no interval.

2. False precision. 0.588033 is written, while $SE = 0.086232$.

3. Rounded numbers are subtracted. 14.17 vs 14.1645 (week 13).

4. Condition not checked. Condition I is violated in 2 of 5 rows in this data.

5. One specification is given. 23 of 48 were significant.

6-10 6. Seed is not written. The range limit floats to 0.044627.

7. Causal language. "increased", "affected" — in observational data (week 13).

8. Two ranges are confused. The prediction and the mean differ by 3.162278 times (week 13).

9. There is no "What was not proven" section.

10. The code is missing or does not work.

Ten points of the report (F205) — keep with you

1 Question — one sentence, in a verifiable form 2 Data — source, volume, method of extraction 3 Refinement — what was extracted and WHY (with numbers) 4 Description — center, spread, shape 5 Model — and why it was chosen 6 Conditions — which ones, how they were tested, result 7 Result — effect + range + [], with units 8 Sensitivity — what happens if the main decision changes 9 What was not proven and why 10 Repetition — code, seed, versions, SI statement

In three sentences

1. There were 48 honest answers to one question. If you do not write down which one you chose and why, your result will not be verified.

2. Reproducibility is a technical form of trust: data + code + seed + versions.

3. A good report says what it doesn't know. This is not a weakness — it's a professional skill.

Next week

Week 15: Project presentations and preparation for the final exam.

Presentation 8 minutes + 4 minutes Q&A. Which of the ten points can fit in 8 minutes — this is also a communication exercise.

Common student mistakes and misconceptions

Common mistakes

1. "Multivariate analysis is testing 48 times, i.e., α -hacking." On the contrary, α -hacking is counting 48 and showing one. Multivariate analysis is showing everything. The difference is not in the counting, but in the transparency.

2. "So none of the results can be trusted." Incorrect conclusion. If 47 out of 48 come out in one direction, this is strong evidence. Measuring sensitivity does not destroy trust, it replaces it.

3. "Seeding removes chance, so the result is fake." Seeding does not remove chance — it allows you to restore it. The seedless lower bound floats to 0.044627; this is not a coincidence, but a lack of documentation.

4. "More room — more precision." When $SE = 0.086232$, the last four rooms in 0.588033 are noise. Extra room does not increase precision, it reduces confidence.

A common mistake

5. "The limitations section is a sign of weakness." The opposite for the reader: the author who stated the limitation himself is more likely to have checked the rest. If someone else finds the limitation — the entire report is in doubt.

6. "The conditions are violated, so the analysis is invalid." No — corrected. In this data, the independence condition is violated in 2 of the 5 rows; corrected with α eff and the conclusion is rewritten. Not a correction, but a correction.

7. "The code written by SI is not my code, so I am not responsible." Your name will be on the report. Recalculate each issue of SI, open each source (F206).

8. "It is possible to calculate with rounded numbers." 44.52 – 30.35 and the difference in full precision are not the same. Rounding is at the end, only in printing.

Summary: three sentences

First There is no single answer to a question — there is a set of reasonable answers.

In our example, there are 48: 23 “significant”, 25 not. Your job is not to find a single answer, but to measure and show how wide the set is.

Second Repeatability is not a virtue, it is a minimum.

Data + code + seed + versions (F204). Without these four, your result is unverifiable; an unverifiable result is an opinion, not a proof. One line of code makes the difference.

Third A good report also tells what is not known.

Effect + interval + conditions + constraints + “what is not proven” (F205). Causal language — only after causal design. Trust is born from transparency, not from determination.

Audience Question

Next Step. Before submitting your report, open the slide “Ten Most Common Report Flaws” and read your text once for each point. The average report contains three of them.

References

Main reading

►Çetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics, Chapter 1 and final chapters — openintro.org/book/ims ►Wickham H., Çetinkaya-Rundel M., Grolemund G. R for Data Science (2nd ed.) — r4ds.hadley.nz, “Communicate” section. Open access.

This week's required reading — both short

►Gelman A., Loken E. (2014). The statistical crisis in science. *American Scientist* 102(6), 460–465. — The phrase “branching path” is from this article; page 6. ►Steege S., Tuerlinckx F., Gelman A., Vanpaemel W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science* 11(5), 702–712. — method of today's section A.

Supplement

►Peng R. (2011). Reproducible research in computational science. *Science* 334(6060), 1226–1227. — 2 pages. ►Wilson G. et al. (2017). Good enough practices in scientific computing. *PLoS Computational Biology* 13(6), e1005510. ►Tuft E. *The Visual Display of Quantitative Information*; Simmons et al. (2011) and Wasserstein & Lazar (2016) — from week 10.